

ELCardioCC 2026 — Execution Masterplan

Version: 1.0 | **Created:** 2026-04-04 | **Competition window:** April 5–20, 2026 (16 days) **Target audience:** 9 MSc AI students ready to execute immediately

1. Executive Summary

What this is: ELCardioCC 2026 is a BioASQ/CLEF shared task where teams automatically assign ICD-10 diagnostic codes to Greek-language cardiology discharge letters.

What you are building: A system that reads a Greek hospital discharge letter and outputs a list of ICD-10 codes describing what the patient was diagnosed with, treated for, or had in their history. You submit predicted codes at the document level.

What success looks like: Highest micro-F1 score on a hidden 500-document test set. Last year's winner hit **F1 = 0.8472**. The organizer baseline was **0.8271**. Beating the baseline is expected; beating the winner is the goal.

Recommended high-level strategy: 1. **Primary path:** Fine-tune Greek-BERT / XLM-RoBERTa for multi-label classification over 115 ICD-10 codes. This is the fastest path to a strong baseline and the backbone of your submission. 2. **Secondary path:** Build a NER → Entity Linking pipeline that extracts medical mentions from text and maps them to ICD-10 codes. Use this as a complementary signal. 3. **Ensemble:** Combine both paths. Submit up to 5 systems — use this to hedge.

Why this strategy: Multi-label classification was the strongest approach last year (F1 0.85). NER+EL provides complementary signal and catches codes the classifier misses. Ensembling both maximizes coverage. You have 16 days — this strategy is implementable in that window.

2. Only The Facts That Matter

Verified Facts

Fact	Source
Training set: 2,500 Greek discharge letters	Dataset page
1,453 documents have both mention-level AND document-level annotations; rest have only document-level	Shared Task page
Test set: 500 documents, released April 5, 2026	Important Dates
Submission deadline: April 20, 2026	Important Dates
Label space: 115 ICD-10 codes	Homepage
Evaluation metric: Micro-F1 (also precision & recall reported)	Shared Task page
Format: JSONL — each line has <code>patient_id</code> , <code>text</code> , <code>document_level_annotations</code> (list of lists), optionally <code>mention_level_annotations</code>	Shared Task page
Submission: ZIP file, up to 5 JSONL files (5 different systems)	Shared Task page
Language: Greek (monolingual)	Dataset page
No registration fee	FAQs

Fact	Source
Paper deadline: May 28, 2026	Important Dates
Data access: Register at BioASQ participants area	FAQs
2025 winner (droidlyx/Fudan): F1 0.8472 (MLC-X) using transformer-based approach	ELCardioCC 2025 page
2025 baseline: Greek-BERT for MLC, mBERT for NER/EL	ELCardioCC 2025 page
2025 dataset was 1,000 train / 500 test — 2026 has 2.5× more training data	Comparison of both editions
Top recognition: Certificates + opportunity to present at CLEF 2026 conference	FAQs

Critical Inference (High Confidence)

Inference	Basis
The "list of lists" annotation format means each document has multiple "annotation groups" — each group lists alternative ICD-10 codes for one clinical finding. Your system must predict at least one code from each group.	Shared Task page description
2026 evaluation is document-level only (no separate NER/EL subtasks visible) — but NER/EL can feed into document-level predictions	Only document-level submission format described for 2026
With 115 codes and 2,500 docs, expect significant class imbalance — some codes appear in hundreds of docs, others in single digits	Standard in ICD coding tasks
Greek-BERT (<code>nlpaueb/bert-base-greek-uncased-v1</code>) is the strongest monolingual Greek model available	HuggingFace , ELCardioCC 2025 results
Dictionary-based ICD-10 matching + LLM achieved F1=0.856 on 2025 EL task — this approach may transfer well	Vassileva et al., arXiv:2509.04868

Open Questions

Question	Impact	Action
Is the 2026 task only MLC (no separate NER/EL subtask)?	Determines if NER/EL is scored separately	Check BioASQ participant area after registration
Exact evaluation script — how are list-of-lists matched?	Critical for threshold tuning	Email dndimitri@csd.auth.gr immediately
Are there new ICD-10 codes in 2026 not in 2025?	Affects transfer from 2025 models	Inspect training data as soon as downloaded
Is external data (e.g., Greek ICD-10 dictionaries) allowed?	Affects dictionary-based approaches	Check rules; assume yes unless prohibited
Time zone for April 20 deadline?	Affects final submission buffer	Assume midnight Greek time (EEST, UTC+3); submit 12h early

3. Problem Breakdown in Plain English

What are the raw documents?

Hospital **discharge letters** — the document a doctor writes when a patient leaves the cardiology department. Written in **Greek**.
Contains: - Why the patient came in (chief complaint) - What was found (diagnosis) - Patient's medical history - Medications prescribed
- Cardiac echo / test results - Treatment plan

These are messy, real clinical notes — abbreviations, inconsistent formatting, medical jargon in Greek.

What is ICD-10 coding?

ICD-10 is the **International Classification of Diseases, 10th revision** — a standardized system where every disease/condition gets a code. Example: - **I21.0** = Acute ST-elevation myocardial infarction of anterior wall (heart attack) - **I50.0** = Congestive heart failure - **I10** = Essential hypertension

In hospitals, trained coders read discharge letters and assign ICD-10 codes. This task automates that process. You have **115 possible codes** in the cardiology domain.

What is the "list of lists" format?

Each document's annotation is structured like this:

```
{  
  "document_level_annotations": [  
    ["I21.0", "I21.1"],  
    ["I50.0"],  
    ["I10"]  
  ]  
}
```

Each inner list represents one clinical finding. If the list has multiple codes (like `["I21.0", "I21.1"]`), it means the annotators considered either code acceptable for that finding. Your system must predict **at least one code from each inner list** to get credit.

What is NER here?

Named Entity Recognition = finding the text spans in the discharge letter that mention diseases, symptoms, drugs, procedures.
Example: in the Greek text, spotting "οξύ έμφραγμα μυοκαρδίου" (acute myocardial infarction) at character positions 145–172.

What is Entity Linking here?

Entity Linking = taking each entity found by NER and mapping it to the correct ICD-10 code. "οξύ έμφραγμα μυοκαρδίου" → **I21.0**.

What is Multi-label Classification here?

Treating the whole document as input, and predicting which of the 115 ICD-10 codes apply. No need to find where in the text — just which codes. This is the **primary evaluation target**.

What does explainability mean here?

In 2025, there was an explainability subtask (MLC-X) — justify which part of the text supports each predicted code. For 2026, **this may not be a separate scored subtask**, but building explainability is still good practice and useful for error analysis.

4. Recommended Winning Strategy

Primary Path: Multi-Label Classification (MLC)

This is your main bet. It directly optimizes the evaluation metric.

- Fine-tune **Greek-BERT** (`nlpau/bert-base-greek-uncased-v1`) or **XLM-RoBERTa-large** for multi-label classification
- Input: full document text (truncated/chunked to fit 512 tokens)
- Output: 115 sigmoid outputs, one per ICD-10 code
- Loss: Binary cross-entropy with class weights for imbalance
- Threshold: Tune per-class thresholds on validation set to maximize micro-F1

Why this wins: Last year's best MLC score (0.8472) beat every other approach. With 2.5× more training data in 2026, expect even better results.

Secondary Path: NER → Entity Linking Pipeline

- Fine-tune Greek-BERT or mBERT for token classification (BIO tagging) on the 1,453 documents with mention-level annotations
- Link extracted entities to ICD-10 codes using:
 - A classifier trained on mention → code mapping
 - Or dictionary matching against Greek ICD-10 terminology + embedding similarity
- Aggregate entity-level codes to document-level predictions

Why this helps: Catches codes the MLC classifier misses, especially rare ones. Provides a complementary signal for ensembling.

Ensemble Strategy

Combine MLC and NER+EL outputs: - Union of predicted codes (boosts recall) - Weighted voting (if confidence scores available) - Simple rule: if either system predicts a code with confidence > threshold, include it

Backup Path (if fine-tuning underperforms)

Dictionary + LLM approach (proven F1=0.856 on 2025 EL data): - Build a Greek ICD-10 dictionary from official WHO translations - Match dictionary terms against document text - Use an LLM (GPT-4 / Claude) for ambiguous cases with few-shot prompting - This requires no training and can be built in 1–2 days

Optional Stretch: Long Document Handling

Discharge letters may exceed 512 tokens. Options: - **Sliding window** with overlapping chunks, aggregate predictions - **Longformer** or **BigBird** variants (if Greek-compatible) - **Hierarchical attention** — encode chunks separately, then aggregate

What NOT to do

- Do NOT build a retrieval-augmented generation system — overkill for this task
- Do NOT spend days on data augmentation before having a working baseline
- Do NOT try to train a model from scratch — fine-tune pretrained models
- Do NOT optimize for explainability at the cost of F1 — F1 is what's scored
- Do NOT build a complex pipeline before having a simple baseline submitted

5. Team Structure and Role Assignment

9-Person Configuration

#	Name	Role	Mission	Key Responsibilities	Deliverables	Dependencies
1	Vasiliki	Team Lead & MLC Architect	Own the primary MLC pipeline end-to-end	Model architecture, training loop, hyperparameter search, final model selection	Working MLC model, validation results, final predictions	Needs clean data from Froso by day 2
2	Stelios	NER+EL Pipeline Lead	Build the complementary NER→EL pipeline	NER model training, entity linking module, code aggregation logic	NER model, EL module, document-level code predictions	Needs mention-level data from Froso by day 2
3	Froso	Data Engineer & EDA Lead	Make data ready and understood	Data loading, cleaning, EDA, splits, label distribution analysis, data loaders	Clean datasets, EDA report, train/val splits, PyTorch dataloaders	Blocks Vasiliki and Stelios
4	Chalkias	Experiment & Infra Lead	Keep experiments reproducible and fast	GPU setup, experiment tracking (W&B/MLflow), training scripts, model checkpointing	Running infra, experiment dashboard, reproducible training configs	Start immediately, blocks nobody
5	Strafiotis	Evaluation & Submission Lead	Own evaluation metrics and submission packaging	Implement micro-F1 evaluation, threshold tuning, submission format validation, final ZIP packaging	Evaluation script, threshold optimizer, submission validator	Needs evaluation format clarity (email organizers day 1)
6	Glikeria	Greek NLP & Dictionary Specialist	Handle Greek language-specific challenges	Greek ICD-10 dictionary, text preprocessing, Greek tokenization analysis, dictionary-based baseline	Greek ICD-10 lookup table, preprocessing pipeline, dictionary baseline predictions	Can start immediately in parallel
7	Stanimeros	Ensemble & Error Analysis	Maximize F1 through combination strategies	Ensemble methods, error analysis, confusion matrices, failure categorization, ablation studies	Ensemble pipeline, error analysis report, improved predictions	Needs predictions from Vasiliki & Stelios (day 8+)
8	Panagiotis	Backup Systems & LLM Experiments	Explore LLM-based approaches as hedge	Few-shot prompting with GPT-4/Claude, zero-shot classification, LLM-based entity linking	LLM-based predictions, cost analysis, comparison with fine-tuned models	Can start immediately in parallel
9	Dimitris	Second MLC Track & Long-Doc Specialist	Run parallel MLC experiments + solve long document problem	Train XLM-R-large MLC in parallel with Vasiliki's Greek-BERT, implement sliding window for long docs, data augmentation for rare codes	XLM-R-large MLC model, sliding window implementation, rare-code augmentation results	Needs clean data from Froso by day 2; GPU from Chalkias

Daily Output Expectations

Role	Daily Deliverable
Vasiliki	Training logs, validation F1, model iterations pushed to shared repo
Stelios	NER training metrics, linked entity counts, pipeline output samples
Froso	Data artifacts (day 1–3), then support data issues for others
Chalkias	Infra uptime, experiment logs, GPU utilization reports
Strafiotis	Evaluation results on every new model output, leaderboard tracking
Glikeria	Dictionary coverage stats, preprocessing quality checks

Role	Daily Deliverable
Stanimeros	Error breakdowns once models produce output, ensemble results from day 8+
Panagiotis	LLM experiment results, prompt iterations, cost tracking
Dimitris	XLM-R training logs, sliding window experiment results, rare-code analysis

Scaling to 14–20 People (If Available)

Add these roles: - **1 additional MLC experimenter** — try mBERT or other architectures in parallel - **1 additional NER person** — help Stelios with mention-level annotation processing - **1 data augmentation specialist** — back-translation, synonym replacement for rare codes - **1 literature reviewer** — continuously scan for relevant papers/code - **1–2 additional ensemble/ablation people** — systematic hyperparameter sweeps

6. Workstream Breakdown

WS1: Data Preparation & EDA

Aspect	Detail
Goal	Clean, understood, split data ready for all downstream tasks
Owner	Froso
Inputs	Raw JSONL from BioASQ
Outputs	Train/val split (80/20 stratified), label frequency table, EDA notebook, PyTorch/HF datasets
Tools	Python, pandas, matplotlib, seaborn, HuggingFace datasets
Fastest path	Load JSONL → compute label stats → stratified split → save → done
Risks	Encoding issues with Greek text; inconsistent annotations
Done when	All team members can <code>load_dataset()</code> and see clean data with labels

WS2: Multi-Label Classification

Aspect	Detail
Goal	Best possible micro-F1 on document-level ICD-10 prediction
Owner	Vasiliki (supported by Chalkias for infra)
Inputs	Clean train/val splits from Froso
Outputs	Trained model, validation F1, test predictions
Tools	HuggingFace Transformers, PyTorch, Greek-BERT, XLM-RoBERTa
Fastest path	<code>AutoModelForSequenceClassification</code> with <code>problem_type="multi_label_classification"</code> , 115 labels, BCE loss
Risks	512-token limit truncates long documents; class imbalance crushes rare codes
Done when	Val micro-F1 ≥ 0.80

WS3: NER + Entity Linking Pipeline

Aspect	Detail
Goal	Extract medical entities and map to ICD-10 codes
Owner	Stelios
Inputs	1,453 mention-annotated documents
Outputs	NER model, EL module, document-level code aggregation
Tools	HuggingFace token classification, spaCy (optional), Greek-BERT
Fastest path	Fine-tune Greek-BERT for token classification with BIO tags → train code classifier on extracted mentions
Risks	Only 1,453 docs have mention annotations; NER errors cascade into EL errors
Done when	Pipeline produces document-level codes; NER F1 ≥ 0.70

WS4: Greek ICD-10 Dictionary & Preprocessing

Aspect	Detail
Goal	Greek-specific NLP resources ready for all teams
Owner	Glikeria
Inputs	Official Greek ICD-10 translation, training data
Outputs	ICD-10 code → Greek description lookup, text preprocessor, dictionary-based baseline
Tools	Python, regex, unicodedata, WHO ICD-10 Greek resources
Fastest path	Download Greek ICD-10 list → build exact/fuzzy matcher → run on training data → measure coverage
Risks	Clinical abbreviations won't match dictionary terms
Done when	Dictionary baseline produces predictions; coverage stats reported

WS5: Evaluation & Submission

Aspect	Detail
Goal	Reliable evaluation pipeline and valid submission packages
Owner	Strafiotis
Inputs	Predictions from all systems, official evaluation format
Outputs	Micro-F1 scores, per-class metrics, threshold optimizer, submission ZIPs
Tools	scikit-learn, custom eval script matching competition format
Fastest path	Implement list-of-lists matching logic → wrap in eval function → test on val set
Risks	Misunderstanding the list-of-lists evaluation → optimizing wrong metric
Done when	Eval script validated against sample; submission format verified

WS6: Ensemble & Final Optimization

Aspect	Detail
Goal	Combine all systems for maximum F1

Aspect	Detail
Owner	Stanimeros
Inputs	Predictions from WS2, WS3, WS4, WS8
Outputs	Ensemble predictions, ablation results
Tools	Python, scikit-learn, custom voting logic
Fastest path	Union of predictions from top-2 systems → tune inclusion threshold
Risks	Ensembling hurts if individual systems are weak
Done when	Ensemble F1 > best individual system F1

WS7: Infrastructure & Experiment Tracking

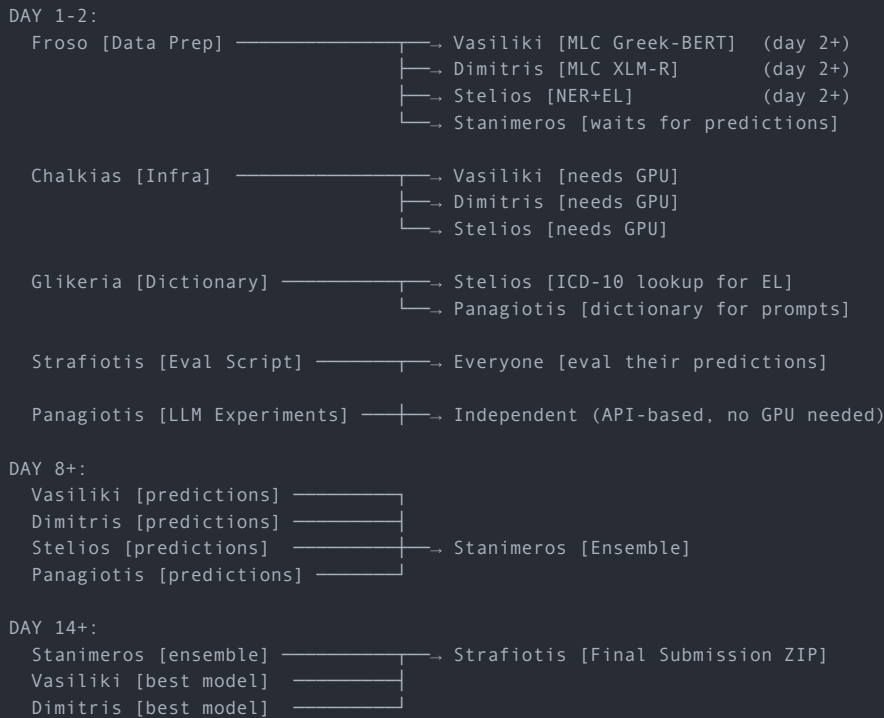
Aspect	Detail
Goal	Everyone can train models and track experiments
Owner	Chalkias
Inputs	Team GPU resources (cloud or university)
Outputs	Running GPU instances, W&B/MLflow project, shared configs
Tools	Google Cloud / AWS / university cluster, W&B, Docker (optional)
Fastest path	Set up shared W&B project → create training script template → distribute
Risks	GPU availability; cost overruns
Done when	Every team member can launch a training run and see results in dashboard

WS8: LLM-Based Approaches

Aspect	Detail
Goal	Explore whether LLMs can match/complement fine-tuned models
Owner	Panagiotis
Inputs	Training data samples, Greek ICD-10 dictionary
Outputs	LLM-based predictions, prompt templates, cost analysis
Tools	OpenAI API / Anthropic API, Python
Fastest path	Few-shot prompt with 5 examples → predict on val set → measure F1
Risks	High API cost; rate limits; may not beat fine-tuned models
Done when	LLM baseline F1 measured; cost per document calculated

7. Dependency Map

Text Dependency Graph



Dependency Table

Task	Blocked By	Blocks	Can Start
Data Prep (Froso)	Nothing	MLC, NER+EL, Ensemble	Immediately
Infra (Chalkias)	Nothing	MLC, NER+EL training	Immediately
Eval Script (Strafiotis)	Nothing	Everyone's evaluation	Immediately
Dictionary (Glikeria)	Nothing	NER+EL, LLM prompts	Immediately
LLM Experiments (Panagiotis)	Nothing	Ensemble	Immediately
MLC Greek-BERT (Vasiliki)	Data Prep, Infra	Ensemble	Day 2
MLC XLM-R + Long-Doc (Dimitris)	Data Prep, Infra	Ensemble	Day 2
NER+EL (Stelios)	Data Prep, Infra	Ensemble	Day 2
Ensemble (Stanimeros)	MLC + NER+EL predictions	Final Submission	Day 8

5 of 9 people can start immediately on day 1. No one should be idle.

8. First 72 Hours Plan

Day 0 — 2026-04-04 (TODAY — Friday)

Everyone: - [] Register on BioASQ: <https://participants-area.bioasq.org/accounts/register/> - [] Register for CLEF 2026: <https://www.clef2026.org/registration/> - [] Join team communication channel (Discord/Slack) - [] Clone shared repo, read this document

Froso (Data): - [] Download ELCardioCC 2026 training data from BioASQ - [] Load JSONL, verify structure, count documents - [] Report: how many docs, how many with mention annotations, label distribution

Chalkias (Infra): - [] Set up shared Git repo with folder structure (see Section 15) - [] Provision GPU access (Google Colab Pro / university cluster / cloud) - [] Set up W&B project, share invite link - [] Create a training script template

Strafiotis (Eval): - [] Email organizers (dndimitri@csd.auth.gr) asking for: evaluation script, list-of-lists matching clarification, time zone for deadline - [] Implement micro-F1 evaluation matching the list-of-lists format - [] Create a dummy submission to validate ZIP format

Glikeria (Dictionary): - [] Find and download Greek ICD-10 code descriptions (WHO, EODY, or extract from training data) - [] Build code → description lookup table for all 115 codes - [] Start dictionary matcher prototype

Panagiotis (LLM): - [] Design few-shot prompt for ICD-10 coding - [] Test on 10 training documents with GPT-4 / Claude - [] Report: rough F1 estimate, cost per document

Vasiliki (MLC), Dimitris (MLC-2) & Stelios (NER): - [] Read ELCardioCC 2025 overview paper: https://ceur-ws.org/Vol-4038/paper_4.pdf - [] Read Enigma team paper: https://ceur-ws.org/Vol-4038/paper_49.pdf - [] Study Greek-BERT usage for the task - [] Prepare model architecture code (can't train without data splits yet) - [] **Dimitris:** Additionally research sliding window and Longformer approaches for long documents

End-of-day checkpoint: - Training data downloaded and loaded ✓ - Label distribution known ✓ - GPU access confirmed ✓ - Repo set up ✓ - Eval script draft ready ✓

Day 1 — 2026-04-05 (Saturday — TEST SET RELEASED)

Froso: - [] Finalize train/val split (80/20, stratified by label frequency) - [] Create PyTorch dataloaders / HuggingFace Dataset objects - [] Publish EDA notebook with: label frequencies, document lengths, token counts per document - [] Download test set as soon as released

Vasiliki (MLC — Greek-BERT track): - [] Start training Greek-BERT multi-label classifier - [] First training run: 3 epochs, $lr=2e-5$, $max_length=512$ - [] Report validation micro-F1 by end of day

Dimitris (MLC — XLM-R track): - [] Start training XLM-RoBERTa-large multi-label classifier in parallel with Vasiliki - [] First training run: 3 epochs, $lr=1e-5$, $max_length=512$ - [] Analyze document length distribution — identify how many docs exceed 512 tokens - [] Report validation micro-F1 by end of day

Stelios (NER): - [] Start training NER model on 1,453 mention-annotated docs - [] Use BIO tagging with Greek-BERT or mBERT - [] Report NER F1 on validation set by end of day

Chalkias: - [] Ensure all training runs are logged to W&B - [] Debug any GPU/memory issues - [] Set up model checkpointing

Strafiotis: - [] Validate eval script against training data (self-eval) - [] Test submission ZIP creation with dummy predictions - [] Prepare threshold tuning script

Glikeria: - [] Run dictionary matcher on validation set - [] Report dictionary baseline F1 - [] Identify which codes are NOT covered by dictionary

Panagiotis: - [] Run LLM on 50 validation documents - [] Report F1, precision, recall - [] Identify which codes LLM handles well vs. poorly

Stanimeros: - [] Begin building ensemble framework (code that takes N prediction files → merged output) - [] Start error analysis framework (code that compares predictions vs. gold, categorizes errors)

End-of-day checkpoint: - First MLC validation F1 number exists ✓ - First NER F1 number exists ✓ - Test set downloaded ✓ - Dictionary baseline F1 known ✓

Warning signs: If MLC F1 < 0.50 on first run, check data loading / label encoding. If NER F1 < 0.40, check BIO tag alignment.

Day 2 — 2026-04-06 (Sunday)

Vasiliki: - [] Iterate Greek-BERT: different learning rates (1e-5, 3e-5), different max_length (256, 512) - [] Try class-weighted BCE loss for imbalanced codes - [] Target: val micro-F1 ≥ 0.70

Dimitris: - [] Iterate XLM-R-large: different learning rates, try with/without class weights - [] If docs > 512 tokens are common, implement sliding window approach - [] Target: val micro-F1 ≥ 0.70

Stelios: - [] Build entity linking module: mention $\rightarrow$ ICD-10 code - [] Aggregate entity codes to document-level predictions - [] Report end-to-end NER $\rightarrow$ EL $\rightarrow$ document F1

Froso: - [] Support anyone with data issues - [] Analyze: which documents are hardest? Which codes are most confused? - [] Create per-code frequency table and share with team

Strafiotis: - [] Run evaluation on all available predictions - [] Create team leaderboard (internal) - [] Begin per-class threshold optimization

Everyone else: Continue assigned workstream.

End-of-day checkpoint: - MLC val F1 ≥ 0.70 ✓ - At least 2 different model architectures tried ✓ - NER $\rightarrow$ EL pipeline produces document-level predictions ✓ - Internal leaderboard exists ✓

Day 3 — 2026-04-07 (Monday)

Vasiliki: - [] Try label description-aware classification (use ICD-10 descriptions as auxiliary input) - [] Hyperparameter sweep on best Greek-BERT config - [] Target: val micro-F1 ≥ 0.78

Dimitris: - [] Finalize sliding window implementation if needed - [] Try data augmentation for rare codes (back-translation, oversampling) - [] Target: XLM-R val micro-F1 ≥ 0.78

Stelios: - [] Improve EL accuracy using Glikeria's dictionary - [] Try combining NER mentions with dictionary matching - [] Target: pipeline document-level F1 ≥ 0.65

Stanimeros: - [] First ensemble attempt: combine MLC + NER+EL outputs - [] Report ensemble F1 vs. individual systems

Panagiotis: - [] If LLM F1 is competitive (≥ 0.70), expand to full validation set - [] If LLM F1 is poor (< 0.60), pivot to supporting Vasiliki on MLC experiments

End-of-day checkpoint: - MLC val F1 ≥ 0.78 ✓ - First ensemble result exists ✓ - Clear picture of which approach is strongest ✓

Warning signs: If best system F1 < 0.65 by end of day 3, schedule emergency team meeting. Consider pivoting to dictionary + LLM approach as primary.

9. Delivery Benchmarks and Confidence Gates

Date	Day	Required Output	Owner	Why It Matters	Recovery if Missed
Apr 4	0	Data downloaded, repo set up, GPUs ready	Froso, Chalkias	Everything depends on this	Cannot proceed — all-hands on setup
Apr 5	1	First MLC F1 number on validation	Vasiliki	Proves pipeline works end-to-end	Debug data loading and model setup urgently
Apr 6	2	MLC val F1 ≥ 0.70	Vasiliki	Confirms viable approach	Try XLM-R-large; check label encoding; increase epochs
Apr 7	3	MLC val F1 ≥ 0.78	Vasiliki	On track to beat 2025 baseline (0.83)	Focus all resources on MLC; drop NER if needed

Date	Day	Required Output	Owner	Why It Matters	Recovery if Missed
Apr 9	5	First ensemble result	Stanimeros	Validates multi-system strategy	If no ensemble gain, focus on best single model
Apr 10	6	MLC val F1 ≥ 0.82	Vasiliki	Matches 2025 baseline	Add class-specific tricks, try curriculum learning
Apr 12	8	ARCHITECTURE FREEZE	Vasiliki	Stop trying new architectures	Only tune hyperparams and thresholds after this point
Apr 14	10	Test set predictions from best 3 systems	All	Dry run of submission	If any system can't produce test predictions, drop it
Apr 16	12	DRY RUN SUBMISSION	Strafiotis	Validates submission format and process	Fix format issues; re-submit
Apr 18	14	FINAL MODEL FREEZE	All	No more training after this	Only threshold tuning and ensemble weights
Apr 19	15	FINAL SUBMISSION READY	Strafiotis	24h buffer before deadline	Last-minute fixes only
Apr 20	16	SUBMIT (12h before deadline)	Strafiotis	Buffer for technical issues	If submission fails, email organizers immediately

Hard Pivot Rules

- **If val F1 < 0.65 by day 3:** Abandon complex approaches. Go all-in on dictionary + LLM baseline.
- **If ensemble doesn't help by day 8:** Submit best single model $\times 5$ with different thresholds.
- **If GPU access fails:** Use Google Colab free tier + smaller models (DistilBERT multilingual).
- **If Greek-BERT underperforms XLM-R:** Switch entirely to XLM-R. Don't run both in production.

10. Weekly Sprint Plan Until Submission

Sprint 1: Foundation (Apr 4–6, Days 0–2)

Goal	Deliverable	Checkpoint
Data ready	Clean splits, dataloaders, EDA	Froso confirms data ready
Infra ready	GPUs, W&B, repo	Chalkias confirms training works
First baselines	MLC F1 ≥ 0.70 , NER F1 ≥ 0.50 , Dictionary baseline	Val scores reported
Eval pipeline	Working micro-F1 scorer + threshold tuner	Strafiotis validates against training data

Sprint 2: Iteration (Apr 7–10, Days 3–6)

Goal	Deliverable	Checkpoint
Strong MLC	Val F1 ≥ 0.82	Vasiliki reports daily
Working NER+EL	End-to-end pipeline F1 ≥ 0.65	Stelios reports daily
First ensemble	Combined F1 > best single	Stanimeros reports
LLM assessment	Keep or kill LLM approach	Panagiotis reports

Sprint 3: Optimization (Apr 11–14, Days 7–10)

Goal	Deliverable	Checkpoint
Architecture freeze (day 8)	Final model architecture locked	No new architectures
Threshold optimization	Per-class optimal thresholds	Strafiotis delivers
Error analysis	Top-10 failure modes documented	Stanimeros delivers
Test predictions v1	All systems produce test outputs	Dry run

Sprint 4: Finalization (Apr 15–20, Days 11–16)

Goal	Deliverable	Checkpoint
Dry run submission (day 12)	Valid ZIP uploaded/validated	Strafiotis confirms
Final tuning (days 12–14)	Threshold and ensemble weight optimization	F1 improvement measured
Model freeze (day 14)	No more training	Hard stop
Final submission (day 15)	ZIP ready	24h before deadline
Submit (day 16, 12h early)	Uploaded to BioASQ	Confirmation received
Buffer	Days 15–16 are pure buffer	Use only if needed

11. Technical Recommendations

Language & Frameworks

Tool	Recommendation	Why
Language	Python 3.10+	Standard for ML
Deep Learning	PyTorch 2.x	Best ecosystem for transformers
Transformers	HuggingFace Transformers 4.x	All models available, Trainer API
Tokenizers	HuggingFace Tokenizers	Fast, handles Greek
Data	HuggingFace Datasets + pandas	Efficient loading + EDA

Models to Try (in priority order)

1. `nlpaueb/bert-base-greek-uncased-v1` — Greek-BERT (best monolingual Greek model)
2. `xlm-roberta-large` — Strong multilingual, 550M params, often beats monolingual
3. `bert-base-multilingual-cased` — mBERT, used in 2025 baselines
4. `xlm-roberta-base` — Lighter multilingual option if GPU-constrained

Greek NLP Resources

Resource	Use
Greek ICD-10 dictionary (~11.5K terms)	Dictionary baseline, EL lookup
<code>unicodedata</code> Python library	Accent stripping for Greek text

Resource	Use
<code>spacy</code> with Greek model (<code>el_core_news_sm</code>)	Basic tokenization, sentence splitting

Experiment Tracking

Tool	Setup
Weights & Biases (W&B)	Free for academics. One shared project. Log: loss, F1, lr, batch_size, model_name
Alternative: MLflow	Self-hosted, no account needed

Other Tools

Purpose	Tool
Annotation inspection	Custom Jupyter notebooks, displaCy for NER
Error analysis	pandas + matplotlib, confusion matrices
Version control	Git + GitHub/GitLab
Reproducibility	<code>requirements.txt</code> , random seed fixing, config files
Code quality	Black formatter, minimal — don't over-engineer

Folder Structure

```

elcardiocc-2026/
├── data/
│   ├── raw/           # Original JSONL files (gitignored)
│   ├── processed/     # Cleaned splits
│   └── external/      # ICD-10 dictionaries, etc.
├── src/
│   ├── data/          # Data loading, preprocessing
│   ├── models/        # Model definitions
│   ├── training/      # Training scripts
│   ├── evaluation/    # Eval scripts, metrics
│   ├── inference/     # Prediction scripts
│   └── ensemble/      # Ensemble logic
├── notebooks/         # EDA, error analysis
├── configs/           # Training configs (YAML)
├── submissions/       # Generated submission ZIPs
├── scripts/           # Utility scripts
├── requirements.txt
└── README.md

```

12. Modelling Strategy Details

Baseline Progression

Stage	Model	Expected F1	Time to Implement
Fast baseline	Greek-BERT + linear head, BCE loss, 512 tokens, 5 epochs	0.65–0.72	3–4 hours
Better v2	XLNet-large, class-weighted BCE, sliding window, 10 epochs	0.78–0.83	1–2 days
Best under time	Ensemble of top-2 models + threshold tuning + NER+EL complement	0.83–0.87	By day 10

Multi-Label Classification Details

```
# Core approach — HuggingFace Trainer
from transformers import AutoModelForSequenceClassification

model = AutoModelForSequenceClassification.from_pretrained(
    "nlpau/bert-base-greek-uncased-v1",
    num_labels=115,
    problem_type="multi_label_classification"
)
# Loss: BCEWithLogitsLoss (built-in)
# Output: 115 sigmoid scores — threshold per class
```

Key tricks: - **Per-class threshold tuning:** Don't use 0.5 globally. Optimize threshold per code on validation set to maximize micro-F1. Use grid search (0.1 to 0.9, step 0.05). - **Class-weighted loss:** Weight rare codes higher. $\text{pos_weight} = (N - \text{pos_count}) / \text{pos_count}$ per code. - **Long document handling:** If mean doc length > 400 tokens, use sliding window with stride 256. Aggregate logits across windows (max or mean pooling). - **Learning rate warmup:** 10% of training steps. - **Gradient accumulation:** If batch size is limited by GPU memory, accumulate over 4–8 steps.

Validation Scheme

- **Stratified k-fold (k=5) on multi-label data:** Use iterative stratification (`skmultilearn.model_selection.iterative_train_test_split`).
- **Primary split:** 80/20 for fast iteration. Use k-fold only for final model selection.
- **Never touch test set for validation.** Only predict test set for submission.

Class Imbalance Handling

1. **Focal loss** instead of BCE — reduces contribution of easy-to-classify codes
2. **Oversampling** documents with rare codes (duplicate them in training set)
3. **Per-class thresholds** — the single most impactful technique
4. **Separate "rare code" classifier** trained only on codes with < 20 occurrences

Ensembling Ideas

Method	How	When
Prediction averaging	Average sigmoid scores from multiple models, then threshold	When you have 2+ MLC models
Union ensemble	Take union of codes from MLC + NER+EL	To boost recall
Stacking	Train a small meta-learner on held-out predictions	Only if you have time (day 10+)
Threshold sweep ensemble	Submit same model with 5 different global thresholds	Easy hedge — use for 5 submission slots

Explainability Layer (Optional, If Time Permits)

- Use attention weights from transformer to highlight supporting text spans
- Or: for each predicted code, find the sentence with highest similarity to the ICD-10 description
- This doesn't affect F1 score but helps with error analysis and the paper

13. Data Understanding Guide

First Things to Count

Metric	Command/Code	Why
Total documents	<code>len(data)</code>	Verify: should be 2,500
Docs with mention annotations	<code>sum(1 for d in data if d.get('mention_level_annotations'))</code>	Should be ~1,453
Unique ICD-10 codes	<code>set(code for doc in data for group in doc['document_level_annotations'] for code in group)</code>	Should be ≤ 115
Codes per document (mean, median, max)	Distribution of <code>len(doc['document_level_annotations'])</code>	Understand label density
Document length (characters, tokens)	<code>len(doc['text'])</code> and tokenizer output	Check if 512 tokens is enough
Label frequency	Count occurrences of each code across all documents	Find rare vs. common codes

Distributions to Visualize

1. **Label frequency histogram** — expect long tail; some codes in 500+ docs, some in < 5
2. **Document length histogram** — check how many exceed 512 tokens
3. **Codes per document histogram** — how many codes does a typical document have?
4. **Code co-occurrence heatmap** — which codes appear together? (top 20 codes)
5. **Mention count per document** — for the 1,453 annotated docs

What to Watch For

- **Noisy labels:** Documents where annotation groups seem contradictory or unlikely
- **Annotation inconsistencies:** Same text span annotated differently across documents
- **Label leakage:** Check if ICD codes appear as literal text in documents (they shouldn't, but check)
- **Very short documents:** May lack enough signal for classification
- **Very long documents:** Need sliding window or truncation strategy
- **Codes with < 5 training examples:** Flag these; may need special handling

Creating a Smart Validation Split

```
from iterstrat.ml_stratifiers import MultilabelStratifiedShuffleSplit

# Create binary label matrix
from sklearn.preprocessing import MultiLabelBinarizer
mlb = MultiLabelBinarizer()
y = mlb.fit_transform([
    [code for group in doc['document_level_annotations'] for code in group]
    for doc in data
])

splitter = MultilabelStratifiedShuffleSplit(n_splits=1, test_size=0.2, random_state=42)
train_idx, val_idx = next(splitter.split(X=range(len(data)), y=y))
```

This ensures every ICD-10 code appears proportionally in both train and validation sets.

14. Concrete Deliverables Per Person

#	Name	Main Ownership	Secondary Support	First Deliverable (Day 0-1)	Ongoing Responsibility	Success Criteria
1	Vasiliki	MLC Greek-BERT track	Ensemble input	Training script + first F1 number	Daily model iterations	Val F1 $\geq$ 0.82 by day 6
2	Stelios	NER+EL pipeline	MLC data augmentation ideas	NER training script + first NER F1	Pipeline improvements	Pipeline doc-level F1 $\geq$ 0.65
3	Froso	Data engineering + EDA	Support everyone's data needs	Clean data splits + EDA notebook	Data issue troubleshooting	All team members have working data
4	Chalkias	Infrastructure + DevOps	Training script templates	GPU access + W&B project + repo	Infra stability	Zero downtime, all runs tracked
5	Strafiotis	Evaluation + Submission	Threshold optimization	Eval script + submission validator	Leaderboard tracking	Valid submission by day 12
6	Glikeria	Greek NLP + Dictionary	Preprocessing for all	ICD-10 dictionary + Greek preprocessor	Dictionary coverage improvements	Dictionary baseline F1 reported
7	Stanimeros	Ensemble + Error analysis	Model debugging support	Ensemble framework code	Error reports + ensemble tuning	Ensemble F1 > best single model
8	Panagiotis	LLM experiments + backup	Any underperforming workflow	LLM prompt + 10-doc test results	LLM iterations or pivot to support	LLM F1 measured; pivot decision by day 3
9	Dimitris	MLC XLM-R track + Long-doc	Rare-code augmentation	XLM-R training script + doc length analysis	Daily model iterations, sliding window	Val F1 $\geq$ 0.82 by day 6

15. Repo and Collaboration Setup

Git Branching Model

```
main (protected — only merged after review)
├─ dev (integration branch — merge here first)
├─ feature/mlc-greek-bert (Vasiliki)
├─ feature/mlc-xlm-roberta (Dimitris)
├─ feature/ner-pipeline (Stelios)
├─ feature/data-prep (Froso)
├─ feature/eval-script (Strafiotis)
├─ feature/dictionary (Glikeria)
├─ feature/ensemble (Stanimeros)
├─ feature/llm-baseline (Panagiotis)
└─ feature/infra (Chalkias)
```

Rules

1. **Never push directly to `main`** — always merge via PR from `dev`
2. **PRs need 1 review** — keep it fast, don't block people
3. **Commit messages:** `[component] what changed` — e.g., `[mlc] add class-weighted BCE loss`
4. **Don't commit data files** — add `data/raw/` to `.gitignore`
5. **Don't commit model checkpoints** — use W&B artifacts or cloud storage
6. **Config files go in `configs/`** — never hardcode hyperparameters

Experiment Logging Rules

Every training run must log to W&B with: - `model_name` (e.g., "greek-bert-uncased") - `learning_rate` - `batch_size` - `max_length` - `num_epochs` - `val_micro_f1` (primary metric) - `val_precision`, `val_recall` - `loss_function` - `notes` (free text: what was different about this run)

Avoiding Chaos Under Time Pressure

- **Daily standup (15 min max):** Each person: what I did, what I'll do, what blocks me
- **Single source of truth:** W&B dashboard for model results, GitHub for code
- **No silent failures:** If something breaks, post immediately in team chat
- **Merge to `dev` at least daily** — don't let branches diverge
- **If in doubt, ask in chat** — don't spend 2 hours stuck silently

16. Risks and Mitigations

Risk	Impact	Probability	Mitigation	Owner	Detect By
GPU access fails/insufficient	Critical	Medium	Have Colab Pro backup; use smaller models	Chalkias	Day 0
Greek-BERT can't handle medical vocab	High	Low	Fallback to XLM-R-large (better OOV handling)	Vasiliki	Day 2
512-token truncation loses critical info	High	Medium	Implement sliding window by day 4	Vasiliki	Day 2 (check doc length stats)
Misunderstand list-of-lists evaluation	Critical	Medium	Email organizers day 0; implement conservatively	Strafiotis	Day 1
Severe class imbalance (codes with <5 examples)	High	High	Per-class thresholds + dictionary fallback for rare codes	Vasiliki, Glikeria	Day 1 (EDA)
Test set distribution differs from train	Medium	Low	Use robust validation; don't overfit to train distribution	Stanimeros	Day 6 (test predictions)
Submission format error	Critical	Low	Dry run submission on day 12; validate ZIP structure	Strafiotis	Day 12
Team member drops out	Medium	Medium	Cross-train: each person documents their work	All	Ongoing
API costs for LLM approach too high	Low	Medium	Cap budget; switch to fine-tuned models if needed	Panagiotis	Day 2
Overengineering wastes time	High	High	Architecture freeze on day 8; no new ideas after day 12	Vasiliki	Day 8

17. Default Decision Rules

If We Disagree, Do This

Situation	Default Decision
Which baseline to trust first?	Greek-BERT MLC. It's closest to what won last year.

Situation	Default Decision
Greek-BERT vs. XLM-R?	Go with whichever has higher val F1 by day 4. If tied, pick XLM-R-large (more capacity).
When to stop tuning hyperparameters?	When the last 3 runs improved F1 by < 0.005 . Move to threshold tuning instead.
When to pivot away from an approach?	If it hasn't reached 80% of the best system's F1 after 3 full days of effort.
When to ensemble?	Always ensemble for final submission. Even a simple union helps.
When to freeze architecture?	Day 8 (April 12) . After this, only tune thresholds and ensemble weights.
When to simplify?	If a component adds < 0.01 F1 improvement, drop it. Complexity is a cost.
How to choose between two models?	Pick the one with higher recall if F1 is within 0.01. Recall is harder to fix than precision.
What to do with a code that has < 5 training examples?	Use dictionary matching for that code. Don't rely on the classifier for it.
Should we write a paper?	Yes, but only after submission. Paper deadline is May 28 — plenty of time.
Should we spend time on explainability?	Only if core F1 is ≥ 0.82 . Otherwise, all effort goes to F1.

18. Exact Submission Deliverable — What You Must Ship

This is the single most important section. If you get everything else wrong but ship this correctly, you still have a result.

What You Submit

A **ZIP file** named after your team (e.g., `our_team.zip`) uploaded to the BioASQ participants area before **April 20, 2026**.

What's Inside the ZIP

Up to **5 JSONL files**, each representing a different system/approach:

```
our_team.zip
├─ mlc_greek_bert.jsonl      # System 1: Greek-BERT MLC
├─ mlc_xlm_roberta.jsonl    # System 2: XLM-R MLC
├─ ner_el_pipeline.jsonl    # System 3: NER+EL pipeline
├─ ensemble_union.jsonl     # System 4: Ensemble (union)
└─ ensemble_weighted.jsonl  # System 5: Ensemble (weighted)
```

Exact Format of Each JSONL File

Each line is a JSON object for one patient/document:

```
{"patient_id": 1234, "document_level_annotations": [{"I21.0"}, {"I50.0"}, {"I50.1"}, {"I10"}]}
{"patient_id": 1235, "document_level_annotations": [{"I25.1"}, {"I10"}, {"E11.9"}]}
```

Rules: - One JSON object per line (no pretty-printing, no newlines within a record) - `patient_id`: integer matching the test set patient IDs - `document_level_annotations`: list of lists — each inner list contains one or more ICD-10 code strings - Every test document (all 500) must have a prediction line - Order of lines does not matter - ICD-10 codes must be valid codes from the 115-code label space

How It Gets Scored

For each document, your predicted `document_level_annotations` are compared against the gold standard. **Micro-F1** across all documents determines your rank. Precision and recall are also reported.

Submission Validation Checklist — Run Before Every Upload

```

import json, zipfile, os

def validate_submission(zip_path, expected_patient_ids, valid_codes):
    """Run this before every submission. All checks must pass."""
    errors = []

    # Check ZIP structure
    with zipfile.ZipFile(zip_path, 'r') as z:
        files = z.namelist()
        jsonl_files = [f for f in files if f.endswith('.jsonl')]

        if len(jsonl_files) == 0:
            errors.append("ERROR: No .jsonl files in ZIP")
        if len(jsonl_files) > 5:
            errors.append(f"ERROR: {len(jsonl_files)} systems - max is 5")

        for fname in jsonl_files:
            with z.open(fname) as f:
                lines = f.read().decode('utf-8').strip().split('\n')
                seen_ids = set()

                for i, line in enumerate(lines):
                    try:
                        record = json.loads(line)
                    except json.JSONDecodeError:
                        errors.append(f"ERROR: {fname} line {i+1} is not valid JSON")
                        continue

                    # Check required fields
                    if 'patient_id' not in record:
                        errors.append(f"ERROR: {fname} line {i+1} missing patient_id")
                    if 'document_level_annotations' not in record:
                        errors.append(f"ERROR: {fname} line {i+1} missing document_level_annotations")
                        continue

                    seen_ids.add(record['patient_id'])

                    # Check annotations structure
                    annots = record['document_level_annotations']
                    if not isinstance(annots, list):
                        errors.append(f"ERROR: {fname} line {i+1} annotations not a list")
                        continue
                    for j, group in enumerate(annots):
                        if not isinstance(group, list) or len(group) == 0:
                            errors.append(f"ERROR: {fname} line {i+1} group {j} not a non-empty list")
                        for code in group:
                            if code not in valid_codes:
                                errors.append(f"WARNING: {fname} line {i+1} code '{code}' not in training label space")

                # Check coverage
                missing = expected_patient_ids - seen_ids
                extra = seen_ids - expected_patient_ids
                if missing:
                    errors.append(f"ERROR: {fname} missing {len(missing)} patient_ids: {list(missing)[:5]}...")
                if extra:
                    errors.append(f"WARNING: {fname} has {len(extra)} unexpected patient_ids")

            print(f" {fname}: {len(lines)} records, {len(seen_ids)} unique patients")

    if errors:
        print("\n--- VALIDATION FAILED ---")
        for e in errors:
            print(f" {e}")
        return False
    else:
        print("\n--- VALIDATION PASSED ---")
        return True

# Usage:
# test_data = [json.loads(l) for l in open('test.jsonl')]
# expected_ids = {d['patient_id'] for d in test_data}
# valid_codes = {code for doc in train_data for group in doc['document_level_annotations'] for code in group}
# validate_submission('our_team.zip', expected_ids, valid_codes)

```

Pre-Submission Smoke Test Protocol

Run this sequence **at least 48 hours before deadline** (by April 18):

1. **Format check:** Run `validate_submission()` above — must pass with zero errors
2. **Self-eval:** Score your predictions against the training set validation split — F1 should match your tracked experiments
3. **Sanity check:** Open the ZIP, manually inspect 5 random lines from each JSONL — do predictions look reasonable?
4. **Size check:** ZIP should be small (< 10 MB for text-only predictions). If it's huge, something is wrong
5. **Upload test:** If the platform allows, do a test upload on April 16 to catch any platform-side issues
6. **Backup:** Keep a copy of the ZIP locally AND in cloud storage

What Kills Submissions (Avoid These)

- Missing patient IDs in predictions (instant recall = 0 for those docs)
- Wrong field names (`annotations` instead of `document_level_annotations`)
- Empty inner lists `[]` instead of `[["I10"]]`
- Submitting `.json` instead of `.jsonl`
- Pretty-printed JSON with newlines (breaks JSONL format)
- Forgetting to include all 500 test documents
- Submitting training set predictions instead of test set (yes, this happens)

19. Immediate Action Checklist

Execute these items **right now**, in order:

- [] **Everyone:** Register at <https://participants-area.bioasq.org/accounts/register/>
- [] **Everyone:** Register at <https://www.cleft2026.org/registration/> (for CLEF 2026)
- [] **Chalkias:** Create GitHub repo with the folder structure from Section 15
- [] **Chalkias:** Share repo invite with all team members
- [] **Chalkias:** Set up W&B project, share invite link in team chat
- [] **Chalkias:** Confirm GPU access (Colab Pro / university cluster / cloud)
- [] **Froso:** Download training data from <https://participants-area.bioasq.org/datasets/> (look for `elcardiocc26`)
- [] **Froso:** Load JSONL and report document count, label count, mention annotation count in team chat
- [] **Strafiotis:** Email dndimitri@csd.auth.gr with subject "ELCardioCC 2026 — Evaluation Clarification" asking:
 - How exactly are list-of-lists predictions matched against gold?
 - Is there a public evaluation script?
 - What time zone is the April 20 deadline?
- [] **Glikeria:** Google "ICD-10 Greek translation" / "ICD-10 ελληνικά" — find and download the official Greek ICD-10 code list
- [] **Glikeria:** Extract the 115 unique codes from training data and map them to descriptions
- [] **Panagiotis:** Write a first LLM prompt for ICD-10 coding; test on 5 training documents
- [] **Vasiliki:** Read https://ceur-ws.org/Vol-4038/paper_4.pdf (ELCardioCC 2025 overview — 14 pages)
- [] **Vasiliki:** Draft the Greek-BERT MLC training script using HuggingFace Trainer API
- [] **Dimitris:** Draft the XLM-RoBERTa-large MLC training script (parallel track)
- [] **Dimitris:** Research sliding window approaches for long documents
- [] **Stelios:** Read https://ceur-ws.org/Vol-4038/paper_49.pdf (Enigma team approach — 15 pages)
- [] **Stelios:** Draft the NER training script with BIO tagging
- [] **Stanimeros:** Set up ensemble framework (takes N prediction files, outputs merged predictions)
- [] **Everyone:** Confirm in team chat that you've read this document and know your role

Test set drops tomorrow (April 5). The clock starts now.

"The best submission is the one that gets submitted. Ship early, iterate fast, ensemble everything."